



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester –Spring 2024

Paper: Solid State Physics-II

Course Code: PHYS-3402

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(15x2=30)

$\sum_{-2}^2 = 1$

- (a) Does Fermi energy depend on the size (or volume) of the conductor? Give formula.
- (b) What is physical meaning of Fermi temperature?
- (c) What is Fermi wave vector? Derive its expression.
- (d) What is meant by energy gap and why such gaps appear due to Bragg's reflection at zone boundary?
- (e) Define Bloch function. What does it represent physically?
- (f) Find an expression for the number of states in a band.
- (g) Distinguish between intrinsic and extrinsic semiconductors.
- (h) How can the effective masses of electrons and holes be determined by the technique of cyclotron frequency?
- (i) What are acceptor and donor states?
- (j) Why the product np is constant at a given temperature in case of semiconductors?
- (k) Describe briefly the intrinsic mobility.
- (l) What is meant by cyclotron frequency? Give its mathematical expression.
- (m) Prove that the Fermi level lies in the middle of the Fermi level.
- (n) What are the failures of free electron model?
- (o) Do electrons have zero kinetic energy at 0 K? If not, why?

Answer the following questions.

Q. 2: (a) Derive expressions for dispersion relations and Fermi energy for free electron Fermi gas in 3-dimensions. (5)

(b) Prove that kinetic energy of three dimensional gas of N free electrons at 0K is $U_e = \frac{3}{5} N \epsilon_F$ (5)

Q. 3: (a) What do you mean by the nearly free electron model, explain briefly. (5)

(b) How can you establish the origin of energy gap using the idea of the standing waves? (5)

Q. 4: (a) Using the semi-classical model, show that the electron moves on a surface of constant energy in k -space. (5)

(b) Define and derive the expression for the effective mass of an electron. (5)

Short Questions

(Group-I)

→ Paper: Spring 2024

Q1) Does fermi energy depend ----- Give formula.

Yes, fermi energy depends on the electron density, which is related to the volume.

→ Formula:

$$\Rightarrow E_F = \frac{\hbar}{2m} \left(3\pi^2 \frac{N}{V} \right)^{2/3}$$

where

- $E_F \Rightarrow$ fermi energy
- $\hbar \Rightarrow$ reduced planck's constant
- $m \Rightarrow e^-$ mass
- $N/V \Rightarrow$ electron density.

So, if no. of electrons per unit volume changes, fermi energy will also change.

Q2) Physical meaning of fermi temperature:

Fermi Temperature is the conceptual temperature according to the fermi energy:

Formula:

$$T_F = \frac{E_F}{k_B}$$

It tells us the temperature at which the thermal energy $k_B T$ is comparable to the fermi energy. For most metals, T_F is very high ($10^4 - 10^5$ K), meaning at room temp e^- are still degenerated and behave quantum mechanically.

Q3) Fermi vector:

Fermi vector k_F represents the maximum wave vector for e^- at $T=0$ K

Expression:

$$k_F = \left(3\pi^2 \frac{N}{V} \right)^{1/3}$$

Derivation sketch:

Total no of e^- = no of state in a fermi sphere.

$$N = 2 \cdot \frac{V}{(2\pi)^3} \cdot \frac{4\pi}{3} k_F^3$$

By solving: $N = \frac{8\pi V}{3(2\pi)^3} \cdot k_F^3 \Rightarrow \frac{8\pi V}{3(2\pi)^3} \cdot k_F^3 = N \Rightarrow \boxed{\frac{V}{3\pi^2} k_F^3 = N}$ (2)

as

$$k_F^3 = \frac{3\pi^2 N}{V} \Rightarrow k_F = \left(\frac{3\pi^2 N}{V} \right)^{1/3}$$

so $k_F = \left(\frac{3\pi^2 N}{V} \right)^{1/3}$ — (A)

is the required expression.

Q4 what is meant by energy gap? ——— boundary?

An energy gap is the range in solids where no e^- states exist.

A Brillouin zone boundaries due to Bragg reflection of e^- (state exists) waves, standing waves are formed. These result in energy splitting (gap) b/w bands. It's a quantum mechanical effect due to the periodic lattice potential.

Q5 Bloch function:

In a periodic potential, e^- wave functions are:

Formula:

$$\psi_k(r) = u_k(r) e^{ik \cdot r}$$

Physical representation:

- $u_k(r)$: function with same periodicity as the crystal lattice.
- Represents the delocalized wave-like behaviour of e^- in a periodic crystal.
- Bloch function explain band formation in solids.

Q6 find an expression for the number of states in a band.

For e^- in 3D space, no of quantum states upto wave vector k :

$$N(k) = 2 \cdot \frac{V}{(2\pi)^3} \cdot \frac{4\pi}{3} k^3$$

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(9) Distinguish

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For a complete band there are two states per atom. so:
 $\Rightarrow$ total states in a band = $2 \times$ Number of unit cells.

(9) Distinguish b/w intrinsic and extrinsic semiconductors.

- Intrinsic
- ① It is a pure semiconductor
 - ② Have equal no of e^- & holes
 - ③ Thermal excitation is source of intrinsic semiconductor.
 - ④ low conductivity

- Extrinsic
- ① It is doped with the impurities
 - ② Majority carriers are dominant
 - ③ Donor/acceptor atoms are source.
 - ④ higher conductivity than intrinsic

(h) How can effective mass ----- frequency?

In magnetic field, charged particles move in circular orbits with cyclotron frequency.

$$\Rightarrow \omega_c = \frac{eB}{m^*}$$

- measure ω_c using resonance methods
- Rearrange to find effective mass:

$$\Rightarrow m^* = \frac{eB}{\omega_c}$$

This is used in cyclotron resonance experiments to determine m^* for e^- & holes in semiconductor.

(i) What are --- donor states?

⊛ Donor State:

From n-type doping. Donor atoms (like P in Si) contribute extra e^-

⊛ Acceptor State:

From p-type doping. Acceptor atom (like B in Si) create holes by accepting e^- .

(j) Why is ----- temperature in semiconductor.

In intrinsic and doped semiconductor at equilibrium

$$np = n_i^2$$

⊛ $n \Rightarrow e^-$ concentration

⊛ $p \Rightarrow$ holes concentration

⊛ $n_i \Rightarrow$ intrinsic carrier concentration

This is known as law of mass action and arises from Fermi Dirac statistics.

(K) Intrinsic mobility-

It is the mobility of charge carriers limited only by lattice scattering, not by impurities.

Given by:

$$\Rightarrow \mu = \frac{e\tau}{m^*}$$

$\Rightarrow e = e^-$ charge

$\Rightarrow \tau =$ mean scattering time

$\Rightarrow m^* =$ effective mass.

(L) what is meant --- expression,

when a charged particle move in a magnetic field, it follows a circular path with some frequency called cyclotron frequency.

$\Rightarrow$ expression

$$\Rightarrow \omega_c = \frac{eB}{m^*}$$

• $\omega_c =$ cyclotron frequency.

• $B =$ magnetic field

• $e =$ charge of particle

• $m^* =$ effective mass.

(m) Prove that --- intrinsic semiconductor.

In an intrinsic semiconductor, the number of e^- = no of holes.

$\Rightarrow$ Using Fermi-Dirac Statistics.

$$n = N_c e^{\frac{-(E_c - E_F)/kT}} \quad , \quad p = N_v e^{\frac{-(E_F - E_v)/kT}}$$

Set $n = p$, solve for E_F .

$$\Rightarrow E_F = \frac{E_c + E_v}{2} + \frac{kT}{2} \ln\left(\frac{N_v}{N_c}\right)$$

if $N_v = N_c$

$$E_F = \frac{E_c + E_v}{2}$$

shows that - fermi level lies at the middle of the band gap.

(m) what are the --- e^- model?

free e^- model (Drude/Lorentz) assumes e^- are free & ignore periodic potential.

→ Failures:

- 1) Can't explain band gaps.
- 2) Doesn't account for electron-lattice interaction
- 3) Fails in insulators and semiconductors
- 4) Can't explain effective mass or mobility differences.

(n) Do ----- why?

No, e^- don't have zero kinetic energy at 0K.

Reason:

Due to the Pauli exclusion principle, e^- occupy energy states up to the Fermi energy at 0K. Only the distribution stops changing; e^- still have energy up to E_F .

Long Question

[all from Kittle:]

Q2: (a) CH # 6 [Pg 137-141]

(b) CH # 6 [Problem Pg 157]

Q3: (a) CH # 7 [164 - 165 Pg]

(b) CH # 7 [Pg 165 - 167]

Q4:

(a) CH # 8 [Pg no 191]

(b) CH # 8 [Pg 197-198]



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Q.1. Answer the following short questions. (10x3=30)

- i) What is Hall Effect? Derive the equation for Hall's coefficient and its significance related to carrier concentration and type of semi-conductor.
- ii) State and prove Bloch theorem and write it in terms of Fourier series.
- iii) Prove that magnitude of Band Gap is equal to the Fourier component of the crystal potential.
- iv) What is a Fermi Sphere & what role does it play in electrical conductivity?
- v) Write down Fermi Dirac Distribution Function and find out probability of occupancy using graphs for the following cases at 0K and at temperatures higher than 0K.
 - a) $E = E_f$
 - b) $E < E_f$
 - c) $E > E_f$
- vi) Prove that magnitude of Band Gap is equal to the Fourier component of the crystal potential.
- vii) What is difference between direct and indirect band gap materials? Draw the optical graphs for direct and indirect gap materials.
- viii) State Wiedman-Franz law and hence define Lorenz Number giving its value in both CGS and SI units.
- ix) What is Matthiessen's Rule and what do we infer from different intercepts of the graph.
- x) What were the success and failures of Free Electron Gas model. Discuss Nearly Free Electron model and origin of band gaps due to Bragg's reflection.

Answer the following questions.

Q2. Derive an expression for intrinsic charge concentration in semi-conductors. Prove that $n_i = n_g/2$. (8+2)

Q3. Solve the Schrodinger equation for the following case

$$\left. \begin{array}{l} V(x) = 0 \\ V(x) = V_0 \end{array} \right\} \begin{array}{l} 0 < x < a \\ -b < x < 0 \end{array}$$

Discuss formation of allowed and forbidden energy bands on the basis of the above results. Also discuss the conditions whether an energy level is discrete or continuous. (10)

Q4. (a) What is meant by effective mass of an electron? Discuss the conditions when it becomes positive, negative and infinite.

(b) What are holes and enlist its various properties as vacant orbitals in semiconductor. (6+4)

SSP-II

PP 2024 (FALL)

Answer the following short questions.

(i)

What is Hall effect? Derive the equations for Hall's coefficient and its significance related to carrier concentration and type of semiconductor?

Hall effect is the electric field developed between two faces of a conductor in the direction $\vec{j} \times \vec{B}$ where $(\vec{j} \times \vec{B})$ is the flow of current across magnetic field $\vec{B}$.

Consider Lorentz force $\vec{F}$ causing the path of e to be deflected towards front face of conductor block when $\vec{B}$ is applied :- $\vec{F}_L = -e\vec{v} \times \vec{B}$

$$\vec{F}_L = -e v B \sin 90^\circ \quad \therefore \vec{v} \perp \vec{B}$$

Similarly, the charges on the opposite face of block produces Hall force :- $\vec{F}_H = e E_H$

For equilibrium state :- $F_L = F_H$

$$e E_H = -e v B$$

where $v = J_x / ne$ $\therefore J_x = nev$ (current density)

Thus, $E_H = - \left(\frac{v J_x}{ne} \right) B$

$E_H = \left(- \frac{1}{ne} \right) J_x B$

where $\left(- \frac{1}{ne} \right) = R_H$ (Hall's coefficient)

$\Rightarrow$ R_H provides the nature of charge carriers are obtained in the conduction.

R_H is directly proportional to charge density.

$\Rightarrow$ R_H 's sign indicates the type of semi-conductor. If it is +ve, it signifies a p-type semi-conductor and if it is -ve, it signifies an n-type semi-conductor.

-(ii)-

State & prove Bloch's theorem and write it in terms of Fourier series?

The eigen function of the wave equation for periodic potential are the product of a plane wave $\exp(i\vec{k} \cdot \vec{r})$ times a function $u_k(\vec{r})$ with the periodicity of crystal lattice.

$$\psi_k(\vec{r}) = u_k(\vec{r}) \exp(i\vec{k} \cdot \vec{r})$$

Consider a one-D linear chain of N atoms:

$$u_k(x) = u_k(x + Na)$$

$$\text{Thus, } \psi_k(x + Na) = u_k(x + Na) e^{ik(x + Na)}$$

$$\therefore \psi_k(x) = u_k(x) e^{ikx}$$

$$\psi_k(x + Na) = u_k(x + Na) e^{ik(x + Na)}$$

$$\therefore \psi_k(x) = u_k(x) e^{ikx}$$

$$\therefore \psi_k(x) = u_k(x) e^{ikx}$$

$\therefore$ Bloch's condition

$$\text{Also, } \psi_k^*(x + Na) \psi_k(x + Na)$$

$$= \psi_k^*(x) e^{-ikNa} \psi_k(x) e^{ikNa}$$

$$= \psi_k^*(x) \psi_k(x)$$

In terms of Fourier series

$$\psi(x) = \sum_k c_k e^{ikx}$$

(iii)

Prove that magnitude of Band gap is equal to the Fourier component of crystal potential?

The energy difference between the two standing waves :-

$$E_g = \int_0^a dx U(x) [|\psi(+)|^2 - |\psi(-)|^2]$$

The normalized standing waves of 1st B.Z.

$$\text{boundary :- } \psi(+) = \sqrt{2} \cos \frac{\pi x}{a} ; \psi(-) = \sqrt{2} \sin \frac{\pi x}{a}$$

Potential energy of an e^- at position x in crystal :-

$$U(x) = U \cos 2\pi x/a$$

Putting the values:-

$$E_g = 2 \int dx U \cos\left(\frac{2\pi x}{a}\right) \left[\cos^2\left(\frac{\pi x}{a}\right) - \sin^2\left(\frac{\pi x}{a}\right) \right]$$

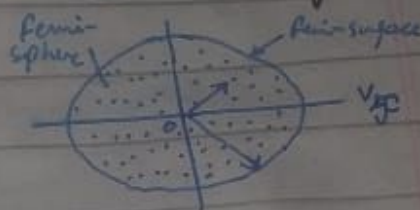
$$E_g = 2 \int dx U \cos^2\left(\frac{2\pi x}{a}\right)$$

$$E_g = U \quad (\text{Proved})$$

(iv)

What is Fermi-Sphere and what is its role in electrical conductivity?

A Fermi-sphere is a simplified model of Fermi-surface.



It shows the distribution of the behaviour of electron in the region of motion.

⇒ At $T=0$ K, the points inside it are all occupied and the points outside it are all empty.

⇒ It defines the boundary between occupied and unoccupied electron states at absolute zero; influencing the number and velocity of electron available for conduction.

(v)

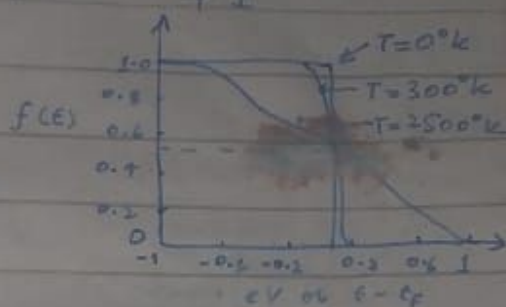
Write Fermi-Dirac distribution func.
Find out prob. of occupancy for the following using graphs at 0K or higher

Fermi-Dirac distribution function gives the probability that an orbital at energy ϵ will be occupied in an ideal electron gas in thermal eq. :-

$$f(\epsilon) = \frac{1}{e^{(\epsilon - \mu)/k_B T} + 1}$$

At $\epsilon = \epsilon_f$

The probability of occupancy is 50% or 0.5.



At $\epsilon < \epsilon_f$

The probability is '1', because at 0K, all energy states below ϵ_f are fully occupied and above states are empty.

At $\epsilon > \epsilon_f$

The probability is '0', because energy states above Fermi level are empty.

(vi)

Same as question 'iv'.

(vii)

Difference between direct and indirect gap materials. Draw optical graphs for them?

Direct band gap

A type of semiconductor material

where the minimum energy level of the conduction band and the maximum energy level of the valence band occur at the same point in the Brillouin Zone is called direct energy gap. It has some crystal momentum k .



Indirect and gap:-

A type of semiconductor material where the min. energy level of c.b. and max. energy level of v.b. do not occur at same point. It has different crystal momentums.



(Vici)

State Widman-Franz law and define Lorenz number giving its

value in CG and SI units

The value of a metal's thermal conductivity (κ) to its electrical conductivity (σ) is

Proportional to the absolute temperature (T).

$$\kappa / \sigma = L T$$

$\therefore L = \text{Lorentz number}$

It differs as the nature of the material is a constant 'L'. (proportionality const.)

In SI units :-

$$L = 2.44 \times 10^{-8} \text{ V}^2 / \text{K}^2$$

In CG units :-

$$L = 2.44 \times 10^{-8} \text{ V}^2 / \text{K}^2$$

Wiedemann-Franz Law

(ix) Wiedemann-Franz Law

What is Matthiessen's Rule and

what do you infer from different intercepts of graphs?

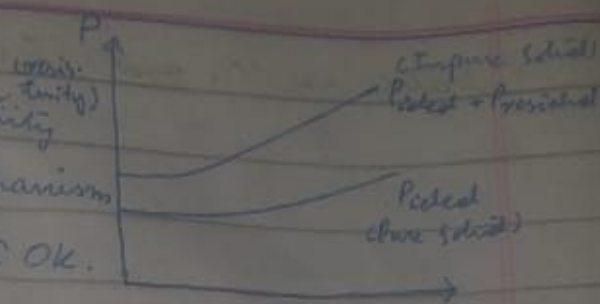
It states that "the sum of resistivity of crystalline metal due to thermal vibrations of the lattice and resistivity due to imperfections in the crystal is the total resistivity of that metal."

$$\rho = \rho_{th} + \rho_{def} + \rho_{imp}$$

$\therefore$ where ' ρ_{def} ' is due to crystal's defects

The graph is as follows:-

The Y-intercept represents the resistivity due to scattering mechanisms other than the phonons at 0K.



And if X-intercept exists, it indicates the temperature at which resistivity due to scattering becomes zero. Overall, the slope represents the increasing 'ρ' with the increasing 'T'.

Successes & failures of free electron gas model. Discuss free e^- gas model and origin of band gaps due to Bragg's reflection?

Free electron gas model treats electrons in a metal as gas confined in a volume.

⇒ It successfully explains the electrical and thermal properties of metals and the Wiedemann-Franz law.

⇒ It fails to accurately describe many aspects of metallic behaviour, e.g; specific heat of metals at low 'T', their magnetic properties, band structure of solids, Hall effect which results in neglecting the quantum

- mechanical nature of e and periodic potential of ion lattice.

Nearly free-electron gas model:-

- It is the quantum mechanical model that describes the behavior of e s in a solid; building upon the free electron model.
- It treats e s as the gas of interacting particles moving freely within a solid.
- In this, the periodic potential arises from the electrostatic interaction between electrons & the positive ions within the solid.

Origin of band gaps due to:-

Bragg's reflection:-

The forbidden energy ranges in materials originate from the interaction of e s waves with the periodic potential of crystal lattice. These interactions reflect electron waves when they satisfy the conditions related to crystal's lattice spacing called Bragg's reflection; which leads to the creation of standing waves and hinders e propagation resulting in band gaps.

Q#02

expression for intrinsic --- $E_f = E_g/2$.

Pg # 205 + this prove, below.

In intrinsic semiconductors, the Fermi level lies in the middle of the band gap!

$$E_F = \frac{E_c + E_v}{2} + \frac{kT}{2} \ln\left(\frac{N_v}{N_c}\right)$$

If $N_c = N_v$, then,

$$E_F = \frac{E_c + E_v}{2}$$

$$E_F = E_v + \frac{E_g}{2}$$

Thus, the Fermi level lies halfway between the conduction and valance bands.

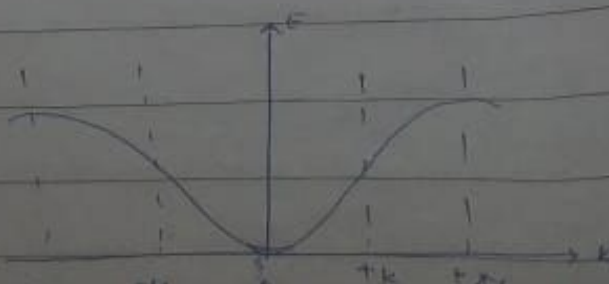
$$E_F = \frac{E_g}{2} \text{ (measured from the valance band)}$$

Q#04

Effective mass of an electron, conditions when it becomes +ve, -ve & infinite.

Pg # 197 + below conditions

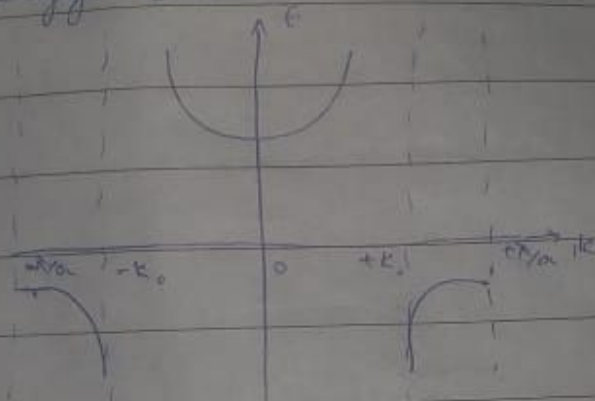
- The effective mass m^* is +ve in the 1st lower half of the energy band.



Date: 1/20

Day:

- The effective mass m^* is -ive in the upper half of the energy band.



- The m^* becomes infinite at the inflexion points ($\pm k_0$) of the $E-k$ curve.

$$m^* = \hbar^2 \frac{1}{\left(\frac{d^2 E}{dk^2} \right)}$$

$$= \hbar^2 \left(\frac{1}{0} \right) = \infty$$

↳ The physical meaning of +ive, -ive, light and heavy mass:

Suppose that $\vec{E}$ is applied on a material. Under the effect of the electric field battery, an electron will try to go toward the positive plate of the battery. If in the path of the electron toward the +ive path, there are many

electrons negatively charged ions arising from the chemical structure of the material. The electron will experience a force of repulsion. Under this force of repulsion, the motion of the electron will slow down or it may or not move at all.

- In case of slow motion, the electron will be said to have heavy mass.
- In case of no motion, the electron will be said to have an infinite mass.

↳ Negative mass?

The electron have -ive mass only when its energy is higher than the inflexion points $(\pm k_0)$. It means the electron with -ive mass are found only near the top of an energy band.

Suppose an electron starts

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wave vector increases linearly with time. Until the velocity reaches its maximum value, the electron is accelerated by the field, beyond this point becomes negative in the upper part of the band.

10/ holes, and its properties:

Pg # 194.

This setup forms a **Kronig-Penney model**-like structure used in **Solid State Physics (SSP)** to study **energy band formation** in crystalline solids.

Q.3

1. Schrödinger Equation in Each Region

The time-independent Schrödinger equation is:

$$\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}[E - V(x)]\psi(x) = 0$$

Region I: $(-b < x < 0)$, $V(x) = 0$

$$\frac{d^2\psi_1(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi_1(x) = 0 \Rightarrow \frac{d^2\psi_1}{dx^2} + k^2\psi_1 = 0$$

where $k = \sqrt{\frac{2mE}{\hbar^2}}$

General solution:

$$\psi_1(x) = Ae^{ikx} + Be^{-ikx}$$

Region II: ($0 < x < a$), $V(x) = V$

$$\frac{d^2\psi_2(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi_2(x) = 0$$

Let $q = \sqrt{\frac{2m(E-V)}{\hbar^2}}$. Note:

- If $E > V$, q is real: propagating wave
- If $E < V$, q is imaginary: evanescent (decaying) wave

General solution:

$$\psi_2(x) = Ce^{iqx} + De^{-iqx}$$

2. Boundary Conditions and Bloch's Theorem

Due to periodicity (unit cell: length $a + b$), we use **Bloch's theorem**:

$$\psi(x + a + b) = e^{iKa} \psi(x)$$

We impose:

- Continuity of ψ and $\frac{d\psi}{dx}$ at $x = 0$ and $x = a$
- Apply Bloch condition across unit cell

This leads to the **Kronig-Penney equation** (transcendental equation):

$$\cos(K(a + b)) = \cos(ka) \cos(qb) - \frac{k^2 + q^2}{2kq} \sin(ka) \sin(qb)$$

Here, K is the **crystal momentum** (quasi-momentum) and lies within the **Brillouin zone**.

3. Allowed and Forbidden Energy Bands

- **Allowed Energy Bands:** When RHS of above equation yields a real value with $|\cos(K(a + b))| \leq 1$, the equation has a **real solution** for K , so these **energy values are allowed**.
- **Forbidden Bands (Band Gaps):** When $\text{RHS} > 1$ or < -1 , **no real solution** for K exists $\Rightarrow$ wave function becomes non-physical $\Rightarrow$ energy is **forbidden**.

4. Discrete vs. Continuous Bands

- **Isolated potential wells (large b or large barrier V):**
 - Energy levels are **discrete**, like in quantum wells
 - Weak overlap between adjacent wavefunctions

4. Discrete vs. Continuous Bands

- **Isolated potential wells (large b or large barrier V):**
 - Energy levels are **discrete**, like in quantum wells
 - Weak overlap between adjacent wavefunctions
- **Periodic potentials with small barrier width/height:**
 - Overlapping wavefunctions create **continuous bands**
 - Energy states become **quasi-continuous**

Key conditions:

- **Discrete Bands:** Arise in non-periodic or strongly localized systems
- **Continuous Bands:** Arise in periodic systems due to **wavefunction overlap**